Plane mirrors

antocciphysics | TEACHER KEY | CPA

Learning target: Use the reflection law and locate a virtual image.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 10; 3 course days later.*
Correct speed=2×50/0.5=200 m/s. The student used half the traveled path and gets half the speed.

02. Explain a worked example (worked)
Image behind mirror at equal perpendicular distance; backward extensions are not actual rays behind an opaque mirror.

03. Look and explain (model)
Reflected ray 30° on other side of normal.

04. Build the solution (practice)
2 m behind; 4 m apart.

05. Use the idea (reason)
Reflected rays appear to originate behind mirror; they lack actual convergence there.

06. Choose a method (mixed) *Review: Unit 6, Day 1; 12 course days later.*
Need wave speed; target distance=speed×4/2. The pulse travels to the target and back.

07. Exit check (exit)
45°.

08. STRETCH: Apply it another way (stretch)
Image 1.5 m behind; separation 3 m.

Plane mirrors

antocciphysics | TEACHER KEY | HONORS

Learning target: Use the reflection law and locate a virtual image.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 10; 3 course days later.*
True speed 340 m/s; incorrect speed≈291.4 m/s, biased low.

02. Explain a worked example (worked)
Image behind mirror at equal perpendicular distance; backward extensions are not actual rays behind an opaque mirror.

03. Represent and interpret (model)
Reflected ray 30° on other side of normal.

04. Solve and justify (practice)
70° from normal; 140° between the rays. Angle based on rays drawn away from point of incidence.

05. Analyze the reasoning (reason)
Reflected rays appear to originate behind mirror; they lack actual convergence there.

06. Choose a method (mixed) *Review: Unit 6, Day 1; 12 course days later.*
Second/first range=6/4≈1.5. Delay does not establish mass.

07. Exit check (exit)
Image moves .50 m toward mirror from behind; separation decreases 1.0 m.

08. STRETCH: Apply it another way (stretch)
Image moves toward mirror from behind at 1.2 m/s; separation decreases 2.4 m/s.